LEXICAL ANALYSER PROGRAM

Aim:

To Write a C program to implement a Lexical Analyzer that identifies keywords, identifiers, constants, operators, and special symbols from a given C statement

Algorithm:

1. Start the program.
2. Read a C statement as input.
3. Scan the input character by character.
4. Ignore white spaces.
5. If the character is a letter or _, read the complete word.
6. Check whether the word is a keyword or identifier.
7. If the character is a digit, identify it as a constant.
8. Check for arithmetic and relational/assignment operators.
9. Identify special symbols such as (), {}, [], ;, and ,.
10. If the character does not match any category, identify it as an unknown symbol.
11. Continue until the end of the input.
12. Stop the program.

Pseudocode:

```
START
Read a C statement as input
Set i=0
WHILE input[i] is not end of string
        IF character is a whitespace
           skip it
        ELSE IF character is a letter or '_'
           Read the complete word
           IF word is a keyword
                 Print "keyword"
           ELSE
                 Print "Identifier"
         ELSE IF character is a digit
               Read the complete number
               Print "constant"
         ELSE IF character is +,-,*,/,%
               Print "Arithmetic Operator"
         ELSE IF character is =,<,>
               Check the next character
```

```
            IF it forms a two-character operator
                Print "Realational Operator"
            ELSE
                Print "Operator"
        ELSE IF character is (, ), {, }, [, ], ;, or ,
                Print "Special symbol"
        ELSE
            Print "Unknown symbol"
        Move to the next character
END WHILE
STOP
```

Sample Input:

```
int a = 10;
```

Sample Output:

```
int     Keyword
a       Identifier
=       Operator
10      Constant
;       Special Symbol
```

Result:

Thus, the Lexical Analyzer was successfully implemented in C, and the given C statement was successfully divided into different tokens.